

Smart IoT-Based Fire Detection & Water Extinguishing System

نظام ذكي لاكتشاف الحرائق وإطفائها باستخدام إنترنت الأشياء

— PROJECT IDEA

An **ESP32**-based system that continuously monitors the environment using a flame sensor. When a fire is detected, it automatically triggers an alarm (LED and buzzer) and activates a water pump through a relay to suppress the fire, while sending the system status to a mobile app over Wi-Fi — allowing the user to monitor the system and control the pump remotely.

— OBJECTIVES

- Early detection of fires before they spread.
- Automatic response to limit the spread of fire.
- Remote monitoring and control via a mobile app.
- A low-cost prototype demonstrating IoT in smart safety systems.

— CORE COMPONENTS

ESP32

Flame Sensor

Relay Module

Water Pump

Buzzer

LED

Wi-Fi

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